

Personal Information

Name: Hussain Burhanuddin

Email: hussainburhan.inbox@gmail.com

Phone: +92-332-2568650

Location: Islamabad

Github: <https://github.com/hussainBurhan>

LinkedIn: <https://www.linkedin.com/in/hussainburhanuddin/>

Portfolio: <https://hussain-portfolio-three.vercel.app/>

D.O.B: 29/08/1999

Professional Experience

Company Name: Futurenostics

Role: Associate Software Engineer -> Software Engineer

Location: Islamabad (on-site)

Duration: Apr 2024 - Present

Projects and work:

LunarScan | Full Stack Developer

- **Architected a comprehensive security dashboard** using **React** and **React Charts**, visualizing real-time threat analysis, KPIs, and vulnerability metrics for end-users.
- **Engineered an asynchronous scanning engine** with **FastAPI** and **Nuclei**, leveraging **Python Multi-threading** to execute parallel vulnerability assessments across large-scale web environments.
- **Developed a sophisticated web crawler** integrated with third-party APIs to discover subdomains, providing a holistic calculation of site-wide threat surface areas.
- **Managed full-cycle database operations**, including ERD design, **PostgreSQL** administration, and automated schema migrations using **Alembic** and **SQLAlchemy**.
- **Optimized financial workflows** by resolving critical integration bugs within the **Stripe** payment gateway.

Diversify | Backend & AI Engineer

- **Developed an automated Computer Vision pipeline** using **Pytesseract OCR** to extract data from shipment labels, significantly reducing manual data entry.
- **Integrated a multi-carrier shipping suite** (FedEx, UPS, USPS, etc.) into a unified **FastAPI** backend to streamline logistics and label generation.
- **Designed scalable NoSQL schemas** using **PynamoDB** for **AWS DynamoDB**, managing complex data migrations and high-concurrency requests.
- **Automated logistics reporting** by building **Playwright** scrapers to aggregate data from disparate shipping service portals.
- **Secured user infrastructure** by implementing robust **OAuth/Social Sign-in** authentication flows and managing cloud assets via **AWS Boto3** and **S3**.

Polaris i/o | Full Stack Developer

- **Translated high-fidelity Figma designs** into responsive, modular **React** components using **Material UI (MUI)**.
- **Streamlined frontend state management** by implementing **Redux** and reducers, ensuring data consistency across complex user workflows.
- **Developed polyglot backend services** across **Node.js** and **FastAPI**, utilizing **Prisma ORM** to interface with a **PostgreSQL** database.
- **Standardized API communication** by implementing Data Transfer Objects (DTOs) and building out a comprehensive library of **RESTful APIs**.

Coinbase | Software Engineer

- **Developed an AI-driven conversational bot** using **Google Dialogflow** and **gRPC** to facilitate seamless user interactions for cryptocurrency trading.
- **Engineered low-latency communication protocols** between the bot interface and platform services to ensure real-time transaction updates.

AI Health Writer | Lead AI Engineer (Solo Developer)

- **Architected and delivered a full-stack MVP** for a healthcare research platform, utilizing **FastAPI**, **React**, and a dual-database strategy (**PostgreSQL** and **MongoDB**).
- **Engineered a Multi-Agent System (MAS)** using **LangChain** to automate complex research workflows, including copywriting style imitation, citation verification, and automated research orchestration.
- **Developed a Hybrid RAG pipeline** combining **semantic and keyword search** with **FAISS** vector database and **MinIO** for scalable document storage.
- **Implemented an interactive Chat-based UI** with document tagging capabilities, allowing users to reference specific research segments for context-aware generation.
- **Designed adversarial AI personas**, such as "Devil's Advocate," to critique healthcare claims and provide **confidence scores** based on verified medical literature.
- **Optimized LLM performance** by leveraging **OpenAI embeddings** and managing complex chat session states for long-form research continuity.

Optikka | Software Engineer (Cloud & Computer Vision)

- **Developed a serverless image processing architecture** on **AWS** for **ESPN**, automating the generation of athlete multi-shot sequences from source imagery.
- **Integrated Computer Vision models** using **OpenCV** and **MediaPipe** to perform high-precision body tracking and image manipulation.
- **Deployed scalable compute solutions** using **AWS Lambda**, **Fargate**, and **EC2**, optimizing resource allocation for high-traffic media processing.
- **Implemented Generative AI workflows** using models like **DALL-E** and **Nano Banana** to dynamically generate and replace athlete backgrounds for broadcast-ready assets.
- **Established robust DevOps practices** by building **CI/CD pipelines** via **GitHub Actions**, ensuring seamless deployment of containerized vision services to **AWS S3** and cloud environments.

AEDI | AI Engineer (Geo-Spatial)

- **Developed a Geo-Spatial RAG pipeline** to automate the retrieval of complex zoning bylaws and land parcel regulations using **PostgreSQL** and **PostGIS**.
- **Engineered a feasibility report generator** using **LangChain** and **GPT-4**, providing automated land usage and construction insights based on spatial data queries.
- **Implemented robust ETL processes** for large-scale geo-spatial datasets, ensuring high-fidelity data was indexed within a **FAISS** vector database.
- **Streamlined urban planning workflows** by translating raw spatial coordinates into human-readable regulatory compliance reports.

Salesforce MCP Server | AI Tools Developer

- **Architected a Model Context Protocol (MCP) server** using **Claude Code**, enabling seamless integration between the Claude Desktop environment and Salesforce CRM.
- **Empowered users to execute CRUD operations** on Salesforce data directly through natural language, significantly reducing time spent on manual record updates.

Admin Bot | AI Automation Engineer

- **Designed a custom internal Admin Agent** using **n8n** and **Google Gemini**, allowing team members to manage system configurations via conversational commands.
- **Built a centralized React-based dashboard** to monitor agent activities and provide a visual interface for complex system queries.
- **Automated internal DevOps tasks** by allowing the bot to securely query, update, and insert configuration data into protected environments.

Lead Generation Automation | Workflow Engineer

- **Engineered an end-to-end lead generation engine** in **n8n**, automating data extraction from **LinkedIn** and company profiles via **Apify**.
- **Integrated a multi-tool stack** including **Airtable** for lead management and **Apollo.io** for real-time contact and email verification.
- **Automated personalized cold outreach** by triggering email sequences based on dynamic lead criteria, increasing the sales pipeline efficiency.

RepairItPro | Machine Learning Engineer

- **Developed an automated invoice processing system** using **Pytesseract OCR** and **Scikit-learn** to extract and categorize billing data.
- **Trained a custom ML model** to map unstructured text from diverse invoice layouts into structured database fields with high accuracy.

Company Name: Techbaltoro

Role: AI Software Engineer

Location: Islamabad, Hybrid

Duration: Dec 2024 - Present

Projects and work:

ecrew.ai | Machine Learning & Full Stack Engineer

- **Engineered a high-performance sales prediction engine** for the Amazon marketplace using **XGBoost** and **LightGBM**, achieving precise forecasting of product demand and revenue.
- **Implemented Explainable AI (XAI)** using **SHAP** values to provide transparency into model features, helping stakeholders understand the "why" behind sales predictions.
- **Established a robust MLOps lifecycle** using **MLflow** to track experiments, log performance metrics, and manage model artifacts for reproducible research.
- **Architected an asynchronous backend** with **FastAPI**, **Celery**, and **Redis** for efficient processing of long-running background tasks and data caching.
- **Built a responsive analytics dashboard** in **React** using **React Charts**, integrating the **Keepa API** to visualize historical pricing and sales rank data.
- **Managed a scalable cloud infrastructure** with a **Supabase (PostgreSQL)** database, **MinIO** for object storage, and automated deployments via **Railway** and **Vercel**.

Returns Tracker | Lead Developer (Mobile & Backend)

- **Developed a cross-platform mobile application** using **Flutter** and **FastAPI** to streamline the management of Amazon customer returns.
- **Integrated LPN (License Plate Number) scanning** functionality to enable rapid identification and tracking of returned inventory within fulfillment workflows.
- **Automated product data retrieval** via **Keepa API** and built a seamless **Google Sheets integration** for real-time return logging and user reporting.

FBA Calculator | Full Stack Extension Developer

- **Developed a browser-based profitability tool** using **React** and **FastAPI** to calculate real-time Amazon FBA margins, fees, and ROI.
- **Optimized high-volume data mining** by utilizing **Google BigQuery** as the primary data warehouse to analyze and serve competitive marketplace insights.

Company Name: Convergent Business technologies

Role: Data Engineer

Location: Islamabad

Duration: Apr 2024 - Dec 2024

Projects and work:

CGAP | Associate Data Analyst & Developer

- **Completed an intensive 6-month technical fellowship** focusing on advanced analytics, software engineering principles, and business intelligence.
- **Mastered core computer science and statistical concepts**, including memory management and algorithms via **Harvard's CS50** and predictive modeling through **Introduction to Statistical Learning (ISLR)**.
- **Specialized in Business Intelligence and SQL**, developing proficiency in complex query optimization and data modeling for large-scale relational databases.
- **Applied best practices in software craft** by implementing principles from the *Code Quality Handbook* and managing project lifecycles using the **Scrum** framework.
- **Designed high-impact data visualizations** based on Stephen Few's principles, utilizing **Tableau** to translate complex datasets into actionable executive insights.

Pepsi | Data Analyst (Contract/Project)

- **Engineered automated data validation pipelines** using **Python**, reducing manual audit time and increasing the accuracy of large-scale supply chain datasets.
- **Managed complex ETL processes** using **MS Excel** and Python to transform raw data into structured formats for corporate reporting.
- **Ensured data integrity** for enterprise-level inventory and sales records, identifying and resolving discrepancies within high-volume data streams.

Company Name: Oxbridge Innovative solution

Role: Design Engineer

Location: Islamabad

Duration: Apr 2022 - Apr -2024

Projects and work:

Oxbridge | Web Developer & Administrator

- **Managed and optimized high-traffic WordPress ecosystems** for *Oxbridge* and *Chotay Karsaaz*, ensuring 99.9% uptime and seamless user experiences.
- **Redesigned site layouts and UI components** to improve brand consistency and mobile responsiveness across multiple web properties.
- **Architected e-commerce workflows** using **WooCommerce**, managing product catalogs, payment gateway configurations, and secure checkout optimizations.
- **Performed routine security audits** and performance tuning, significantly improving page load speeds and SEO rankings.

Python Workshop | Lead Instructor

- **Designed and executed a comprehensive 2-month Python curriculum**, guiding participants from core syntax to intermediate software development concepts.
- **Facilitated hands-on coding labs**, mentoring students on best practices in debugging, modular programming, and data structures.

- **Evaluated student progress** through project-based assessments, fostering a collaborative learning environment for aspiring developers.

Education

Bachelor's of Engineering Mechatronics

Air University, Islamabad

Duration: 2019 - 2023

CGPA: 3.38

A levels

Southshore College, Karachi

Duration: 2020 - 2022

O levels

Avicenna, Karachi

Duration: 2017 - 2020

Certifications

- **Multi Agent Systems with CrewAI** - DeepLearningai
- **Retrieval Augmented Generation (RAG)** - DeepLearningai
- **1 Million AI Prompts** - Dubai Future Foundation
- **Supervised Machine Learning: Regression and Classification** - DeepLearningAI
- **British Airways - Data Science Job Simulation** - Forage
- **Cognizant - Artificial Intelligence Job Simulation** - Forage
- **NLP in Python Natural Language** - MindLuster
- **Attention in Transformers: Concepts and Code in PyTorch** - DeepLearningAI
- **JavaScript Algorithms and Data Structures** - FreeCodeCamp
- **Responsive Web Design** - FreeCodeCamp

Programs

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PIAIC | Agentic AI Specialist (Certification & Training)

- **Mastered the development of autonomous agents** using the **OpenAI SDK** and **Gemini API**, focusing on function calling, multi-agent orchestration, and memory management.
- **Adopted Spec-Driven Development (SDD)** methodologies using **Spec-kit** and **Gemini CLI** to create deterministic, high-quality AI applications by separating technical specifications from implementation.
- **Certified in Core Python (Level 1)**, demonstrating proficiency in data structures, modular programming, and asynchronous execution patterns.
- **Developed agentic workflows** that leverage **Gemini's multimodal capabilities** and command-line tools for automated task execution and complex problem-solving.